

Ivalin Ivanov

DevOps Engineer | CKA | 15+ years in IT

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Professional Overview

DevOps Engineer (CKA) with 15+ years in IT infrastructure, networking and Linux. I design and run a 5-node highly available Kubernetes (k3s) cluster: 3-member etcd with tested failover, about 50 pods across 15 namespaces, public services through Cloudflare Tunnel with SSO, administration over Tailscale, monitoring with Prometheus, Grafana and Loki, and multi-arch CI/CD with GitHub Actions. I debug production issues down to the runtime: profiling the k3s control plane cut its CPU use from 104% to about 10%. Background as a Windows sysadmin at Siemens IT Solutions and Services in the US, networking in Australia, and since 2016 the sole technical person for a small-business IT company with 200+ customers.

Technical Skills

Container Orchestration: Kubernetes (CKA): k3s and kubeadm, HA control plane (etcd quorum), Helm, Ingress (Traefik) and Gateway API, Services and MetalLB, storage (PV/PVC, StorageClass), HPA, CNI (Flannel, Cilium), RBAC, NetworkPolicy, GPU scheduling (NVIDIA device plugin)

Containers and CI/CD: Docker, containerd, buildx multi-arch builds (amd64 + arm64), GitHub Actions, GHCR, Gitea registry; Jenkins (training)

Infrastructure as Code: Terraform, Ansible

Linux and Scripting: Linux system administration (Ubuntu Server, Debian, Fedora, postmarketOS on ARM), systemd services, networking and troubleshooting, Bash, YAML

Monitoring: Prometheus, Alertmanager, Grafana, Loki, Uptime Kuma

Networking and Security: Cloudflare Tunnel and Access SSO (no open inbound ports), Tailscale VPN, SSH hardening with fail2ban, secrets management, network isolation for untrusted devices, Cisco routers

AI-Assisted Engineering: Claude Code CLI for feature development, infrastructure changes, troubleshooting, documentation and workflow automation; every change verified on real systems

Homelab Projects

Self-Hosted HA Kubernetes Platform | bare-metal, always-on

Jan 2026 – present

Tech stack: k3s, embedded etcd, MetalLB, Traefik, Cloudflare Tunnel, Tailscale, Prometheus, Grafana, Loki, GitHub Actions

- 5-node k3s cluster: 2 x86 servers and 3 arm64 OnePlus phones on postmarketOS. 3-member etcd quorum using an etcd-only member; failover tested by stopping a server while the API stayed up. About 50 pods across 15 namespaces.
- No inbound ports: public services through Cloudflare Tunnel (2 replicas) behind Access SSO; administration over a Tailscale mesh with redundant subnet routers.
- Monitoring outside the failure domain: Prometheus, Alertmanager and Loki in-cluster, Grafana and Uptime Kuma on a separate host, alerts by SMTP, routing tuned to actionable alerts only.
- Profiled the control plane in production through the kubelet pprof proxy and found a leaked etcd goroutine; the upgrade cut k3s-server CPU from 104% to about 10%.
- Found a backup CronJob that had not completed once in 28 days; rebuilt it with dependencies baked into the image and added job-failure alerting.
- GitHub Actions builds multi-arch images to GHCR; every manifest kept in Git; incidents published as post-mortems.

Self-Hosted Video Surveillance on Kubernetes | GPU inference, IoT containment

2026 – present

Tech stack: Frigate, Kubernetes, NVIDIA device plugin, TensorRT, MQTT, Cloudflare Access

- Frigate NVR on 6 IP cameras with TensorRT object detection on a GTX 1050 Ti. CPU and memory limits plus kubelet reservations ended repeated node lockups.
- Cameras have no route off the LAN; remote viewing only through SSO-gated Cloudflare Access.

GitHub: github.com/ivemcfire/homelab-config **Blog:** ivemcfire.github.io

Work Experience

Systems & Network Engineer | Vividea LTD, Bulgaria

2016 – present

- Sole technical person in a family-owned small-business IT company: 200+ customers and 200+ installations across offices, shops and homes.
- Server and PC support, network setup and on-site troubleshooting for small businesses.

- IP camera systems with local hosting on Blue Iris and Hikvision; next generation (Frigate on Kubernetes, ANNKE cameras, GPU detection) tested in the homelab.

IT Support Technician | [LowCost IT, Brisbane, Australia](#) **2010 – 2012**

- Network setup and Cisco router configuration, hardware repairs and customer support.

Windows System Administrator | [Siemens IT Solutions and Services, Illinois, USA](#) **2007 – 2009**

- Supported client sites Siemens Energy & Automation (Elk Grove) and KPMG (Schaumburg): Windows servers, Active Directory, Exchange mailboxes and backups.
- Created internal technical documentation and step-by-step tutorials for users and the IT team on new systems and practices; prepared management reports and recommended improvements and optimizations to IT operations.
- On-call for after-hours incidents, including a Friday 9 PM server-room failure at a remote site affecting 500+ users.

Education & Certifications

Bachelor's, Informatics and Computer Science | [University of Economics – Varna, Bulgaria](#) **2004**

Certified Kubernetes Administrator (CKA) | [The Linux Foundation](#) **2026**

DevOps and Cloud 2025 | [SoftUni, Bulgaria](#) **2025 – 2026**

Module certificates: Linux System Administration (92%), Containerization and Infrastructure with Docker, Terraform and Ansible (100%), CI/CD & Monitoring (99%). Kubernetes module certificate expected November 2026. Also took the Azure Essentials module.

KubeCraft Career Accelerator | [hands-on Kubernetes career program](#) **Mar 2026 – Mar 2027**

Homelab-first program. Following its content from January 2026 started my homelab; joined the program in March 2026.

MCSE, MCSA, MCP | [Microsoft, via NetXperts, Inc., Des Plaines, IL, USA](#) **2005 – 2006**

One-year program; also completed the Cisco CCNA course (no exam).

Additional Information

Languages: Bulgarian (native), English (fluent, professional use in the US and Australia)

Work authorization: EU citizen

Location: Targovishte, Bulgaria. Available for remote work